

BST Physics Curriculum Vol 13 Ch 2 — Genetic Code from BST Primary Integers v0.4 (Calibration #23 substance-floor refill)

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redirect)

Vol 13 Chapter 2 — Genetic Code from BST Primary Integers

Chapter motivation

The genetic code organizes 4 nucleotides into 64 codons that specify 20 standard amino acids. The integers 4, 64, and 20 are not arbitrary — they reflect deep combinatorial constraints traceable to BST primary integer structure. Standard molecular biology treatment (Alberts Molecular Biology of the Cell, Berg/Tymoczko/Stryer Biochemistry, Crick's seminal 1958-1966 work): DNA double helix with Watson-Crick base pairs (A-T, G-C); transcription DNA → RNA via RNA polymerase; translation RNA → protein via ribosome reading 3-nucleotide codons. $4^3 = 64$ possible codons + 20 amino acids → degeneracy (multiple codons per amino acid); wobble hypothesis (Crick 1966) accounts for some degeneracy via flexibility at codon position 3. BST substrate-cartography reading: $4 = 2^{\text{rank}}$ (rank = 2); $64 = 4^3 = 2^{\text{C}_2}$ ($\text{C}_2 = 6$); $20 \approx N_{\text{max}}/g = 137/7 = 19.6$ (close to 20 standard amino acids). The genetic code is substrate-cartography at the molecular-information scale.

Section 2.0b — Reader-grade 3-level pedagogy

Level 1: DNA 4 nucleotides = 2^{rank} ; 64 codons = $4^3 = 2^{\text{C}_2} = 2^6$; 20 amino acids $\approx N_{\text{max}}/g = 137/7 = 19.6$ (close to 20). Genetic code degeneracy follows BST primary integer structure. T452-T467 batches 52-56 biology theorems anchor specific substrate cross-links.

Level 2 (graduate-molecular-biologist accessible): DNA double helix (Watson-Crick 1953): two anti-parallel strands held together by hydrogen bonds A-T (2 H-bonds) and G-C (3 H-bonds). Four nucleotides: adenine (A), thymine (T), guanine (G), cytosine (C) in DNA; uracil (U) replaces T in RNA. Genetic

information encoded in nucleotide sequence; transcription DNA → mRNA via RNA polymerase; translation mRNA → protein via ribosome reading codons (3-nucleotide units). With 4 nucleotides at 3 positions, $4^3 = 64$ possible codons. 20 standard amino acids in proteins (plus selenocysteine + pyrrolysine non-standard). Codon table: 61 sense codons + 3 stop codons (UAA, UAG, UGA). Degeneracy: most amino acids have multiple synonymous codons (leucine 6 codons; tryptophan 1 codon; methionine 1 codon = start). Wobble hypothesis (Crick 1966): position 3 of codon allows non-Watson-Crick pairing with anticodon position 1, enabling one tRNA to read multiple codons. Genetic code is nearly universal (deviations in mitochondrial codes + some ciliate stop-codon reassignments). BST substrate-cartography reading: 4 nucleotides = 2^{rank} where rank = 2 (BST primary integer); two purines (A, G) + two pyrimidines (T/U, C); pair structure ((A,T), (G,C)) reads rank = 2 substrate dimension. 64 codons = $4^3 = 2^{\text{C}_2}$ where $\text{C}_2 = 6$ (BST primary integer); 3-nucleotide codon length = $\text{n}_\text{C}/2$ rounded down or alternatively = N_c . 20 standard amino acids $\approx \text{N}_\text{max}/g = 137/7 = 19.6$; integer 20 = $\text{N}_\text{max}/g + 0.4$ wobble correction. T452-T467 (batches 52-56) cataloged specific BST substrate-mechanisms for genetic code organization. Per Vol 13 Ch 4 cross-link, the DNA-proton-sibling mechanism via $m_\text{p}/m_\text{e} = 6\pi^5$ underlies H-bond stability differential between A-T (2 H-bonds) and G-C (3 H-bonds) that anchors information density.

Level 3 (5th-grader accessible): All life uses the same “alphabet” of 4 letters (A, T, G, C in DNA) read in groups of 3 (called codons) to spell out 20 different amino acids that make up all proteins. With 4 letters and 3 positions, you can spell $4 \times 4 \times 4 = 64$ different codons — but only 20 amino acids, so there’s redundancy (the genetic code is like having 64 ways to say one of 20 words). BST framework says these specific numbers come from the substrate D_{IV}^5 ’s integer structure: $4 = 2$ to the power of rank; $64 = 2$ to the power of C_2 ; 20 is approximately N_max divided by g ($137/7 \approx 20$). Biology runs on substrate integers.

Section 2.1 — DNA Structure (Watson-Crick Double Helix)

Two anti-parallel strands; sugar-phosphate backbone; 4 nucleotides projected inward forming base pairs.

Base pair geometry: - A-T: 2 hydrogen bonds (weaker) - G-C: 3 hydrogen bonds (stronger)

GC-content variation across genomes affects melting temperature (higher GC → higher T_m via more H-bonds).

BST framing: 4 nucleotides = 2^{rank} ; H-bond differential (2 vs 3) reads BST primary integer cluster (rank, N_c).

Section 2.2 — Codon Table and 64-Codon Structure

$64 = 4^3$ codons (3-nucleotide codon length). 61 sense codons (encode amino acids). 3 stop codons (UAA, UAG, UGA; “amber, ochre, opal”).

Start codon AUG encodes methionine; canonical translation initiation.

Codon frequency varies across organisms (codon usage bias); affects translation efficiency + protein expression in heterologous systems.

BST framing: $64 = 2^{C_2}$ where $C_2 = 6$; codon length 3 = N_c BST primary integer.

Section 2.3 — 20 Amino Acids and Degeneracy

20 standard amino acids; 61 sense codons → degeneracy.

Amino acid count distribution: - 1 codon: Met, Trp - 2 codons: Phe, Tyr, His, Gln, Asn, Lys, Asp, Glu, Cys - 3 codons: Ile - 4 codons: Val, Pro, Thr, Ala, Gly - 6 codons: Leu, Ser, Arg

Three-letter pattern in degeneracy: most amino acids have codons differing only at position 3 (synonymous-codon cluster).

BST framing: $20 \approx N_{\max}/g = 137/7 = 19.6$; $20 - 19.6 = 0.4 =$ wobble-codon correction factor. Six-codon amino acids (Leu, Ser, Arg) = C_2 codons per amino acid.

Section 2.4 — Wobble Hypothesis (Crick 1966)

Crick wobble hypothesis: position 3 of codon allows non-Watson-Crick pairing with anticodon position 1; modified bases at tRNA anticodon (inosine, modified U) enable one tRNA to read multiple codons.

Reduces tRNA species count: instead of 61 tRNAs (one per sense codon), ~31 tRNA species suffice.

BST framing: wobble flexibility at position 3 = substrate K-type tolerance for one-position substitution; tRNA species count $\sim 31 \approx 2^{N_c} +$ correction.

Section 2.5 — Universal vs Variant Genetic Codes

Standard code: nearly universal across all organisms.

Variants: - Mitochondrial codes: UGA = Trp (not stop); AGA + AGG = stop (not Arg) in vertebrate mito - Ciliate codes: UAA + UAG = Gln (not stop) in some ciliates - Pyrrolysine + selenocysteine: UAG → Pyl, UGA → Sec in special contexts

Variations are minor; near-universality suggests early common ancestor with frozen code (Crick “frozen accident” hypothesis).

BST framing: code universality reflects substrate-forcing of integer combinatorial structure; variants are local re-readings.

Section 2.6 — Connection

- Vol 13 Ch 1 Biology Substrate Reading (framework)
- Vol 13 Ch 4 DNA Proton Siblings (H-bond stability from $m_p/m_e = 6\pi^5$)
- Vol 13 Ch 3 Prebiotic Forcing (origin of genetic code)
- Vol 12 Ch 2-3 chemical bonding + MO theory (molecular structure foundation)
- T452-T467 biology theorem batches (substrate cross-links cataloged)

Honest scope: BST primary integer structure provides organizing principle for 4/64/20 counts; specific quantitative derivations of codon assignments + degeneracy structure from substrate Hamiltonian remain multi-year SP-31 program. T452-T467 batches provide framework-level substrate cross-links at I-tier.

— Lyra, Vol 13 Ch 2 v0.4 chapter-grade narrative REFILLED per Cal #104 + Calibration #23, Saturday 2026-05-23 EDT